

# Sourabh Saxena

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## CAREER OBJECTIVE

AI and Machine Learning enthusiast with hands-on experience building full-stack AI applications, LLM-powered systems, and machine learning solutions. Seeking a Software Engineer/AI-ML Internship to apply skills in Python, FastAPI, Data Analytics, and Generative AI while contributing to real-world products.

## TECHNICAL SKILLS

- **Programming:** Data Structures and Algorithms, OOPS, DBMS
- **Languages:** Python, SQL
- **Frameworks:** FastAPI, React, LangChain, TensorFlow, Scikit-Learn, XGBoost, Keras, Streamlit
- **Libraries & Tools:** Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, FAISS, Git, GitHub
- **Skills:** Machine Learning, Deep Learning, NLP, LLMs, RAG, Transfer Learning, EDA, Git, GitHub

## PROJECTS

**DataPilot – AI-Powered Data Analytics Platform** | [Python](#), [FastAPI](#), [LangChain](#), [React](#), [Groq Llama 3.1](#), [Plotly](#), [SQLite](#) | [GitHub](#) | *May 2026*

- Built an AI-powered data analytics platform using FastAPI, React, LangChain, and Groq Llama 3.1 for natural language analysis of CSV/Excel datasets.
- Developed a custom tool-calling agent to automate dataset exploration, Pandas-based analysis, and interactive Plotly visualizations.
- Deployed a full-stack application on Render and Vercel with secure file uploads, session management, and AI-generated business insights.

**VidyaAI – AI-Powered Smart Tutoring System** | [Python](#), [FastAPI](#), [FAISS](#), [LLMs](#) | [GitHub](#) | *Mar 2026*

- Built an AI tutoring system enabling PDF-based Q&A using LLMs (LLaMA 3.1 via Groq) and embeddings for accurate, textbook-grounded responses.
- Designed a Context Pruning pipeline with FAISS + cosine similarity to retrieve relevant content, reducing latency and cost while maintaining answer quality.
- Collaborated on developing and deploying a full-stack application using React (frontend) and FastAPI (backend), contributing to backend logic and API development.

**Plant Disease Detection Using Transfer Learning** | [TensorFlow](#), [Keras](#), [EfficientNetB0](#), [Streamlit](#) | [GitHub](#) | *Mar 2026*

- Built a 38-class plant disease classifier (54K+ images) using EfficientNetB0 fine-tuning; improved baseline CNN from 94.78% → 98.04% accuracy.
- Achieved 0.98 Macro F1-score using augmentation and regularization for balanced multi-class performance.
- Implemented a modular end-to-end pipeline for preprocessing, training, evaluation, and optimized inference.

## INTERNSHIP

**AI-ML Virtual Intern — EduSkills Foundation (Oct 2025 - Dec 2025) 10-Week | Remote**

- Worked on data preprocessing, exploratory data analysis (EDA), and feature engineering using Python.
- Built and evaluated basic machine learning models using Scikit-learn.
- Performed data analysis using Pandas and NumPy to extract insights from datasets.
- Gained practical exposure to end-to-end ML pipelines and real-world problem-solving approaches.

## EDUCATION

|                                                               |                             |
|---------------------------------------------------------------|-----------------------------|
| <b>IMS Engineering College</b><br>B.Tech ( CSE AI-ML ) - 8.75 | <i>Oct 2023 - July 2027</i> |
| <b>Krishna Vidya Niketan</b><br>High School - 83.83%          | <i>Apr 2022- Apr 2023</i>   |
| <b>Krishna Vidya Niketan</b><br>Secondary School- 90%         | <i>Apr 2020 - May 2021</i>  |

## ACHIEVEMENTS/CERTIFICATIONS

- **Finalist – Meta PyTorch OpenEnv Hackathon x Scaler (2026):** Selected among 52,000+ participants; advanced to the final stage
- Solved 350+ algorithmic problems on LeetCode covering DSA, arrays, strings, and greedy techniques
- **AWS Academy – Generative AI Foundations (2026)** [Certificate](#)
- **AWS Cloud Technical Essentials – Coursera (2026)** [Certificate](#)
- **Data Science & AI 2026 – From Python To Gen AI:** Udemy (2026) [Certificate](#)
- **Prompt Engineer for Developer:** Udemy (2025) [Certificate](#)
- **ChatGPT for Everyone :GUVI (2025)** [Certificate](#)
- **Innovation with Google Cloud AI (Introduction):** Simplilearn (2025)